PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

Juntando técnicas

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Leibniz Institute of Plant Genetics and Crop Plant Research



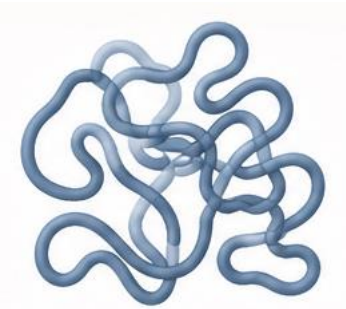
26 de outubro de 2026

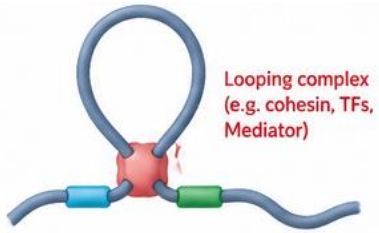
In Silico Genebank Proteomics

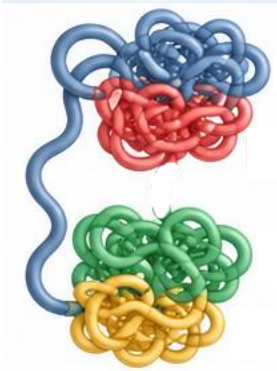
IPK LEIBNIZ-INSTITUT

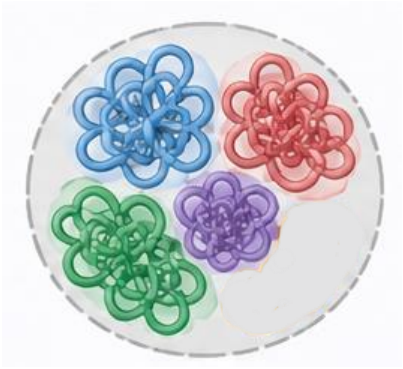
Exercício – combinar a conformação e o mapa de contatos

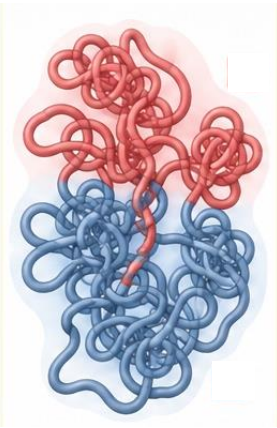
*Feito com IA



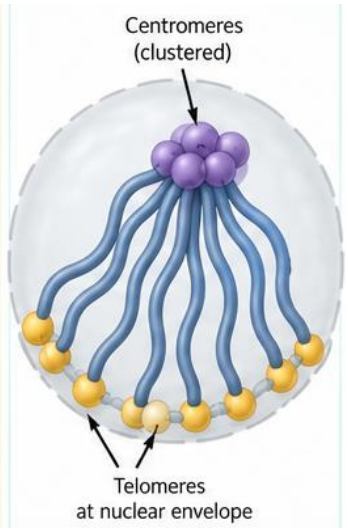


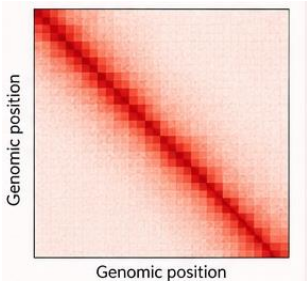


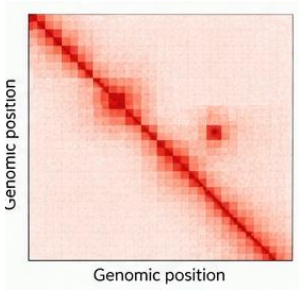


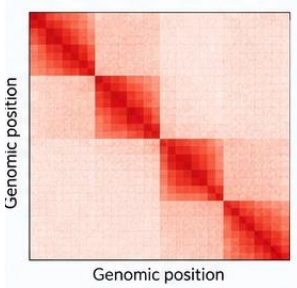


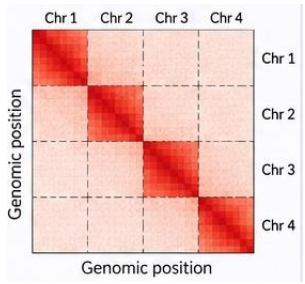


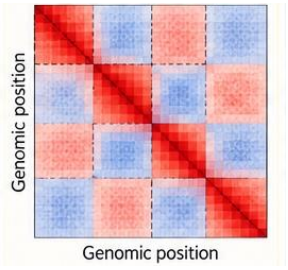


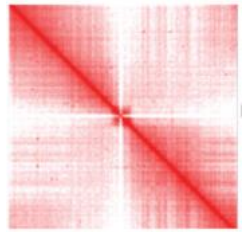


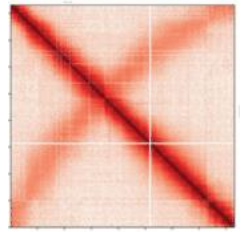




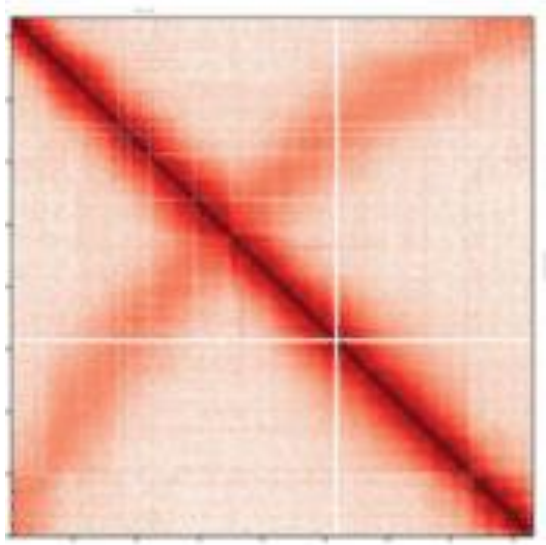


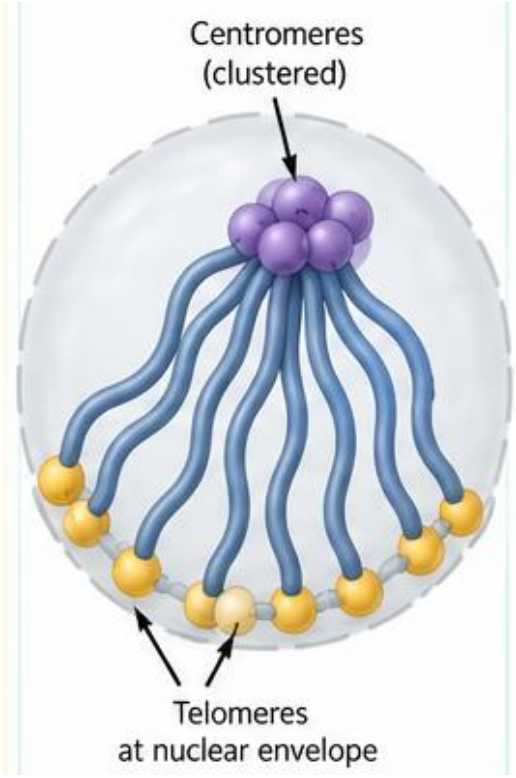


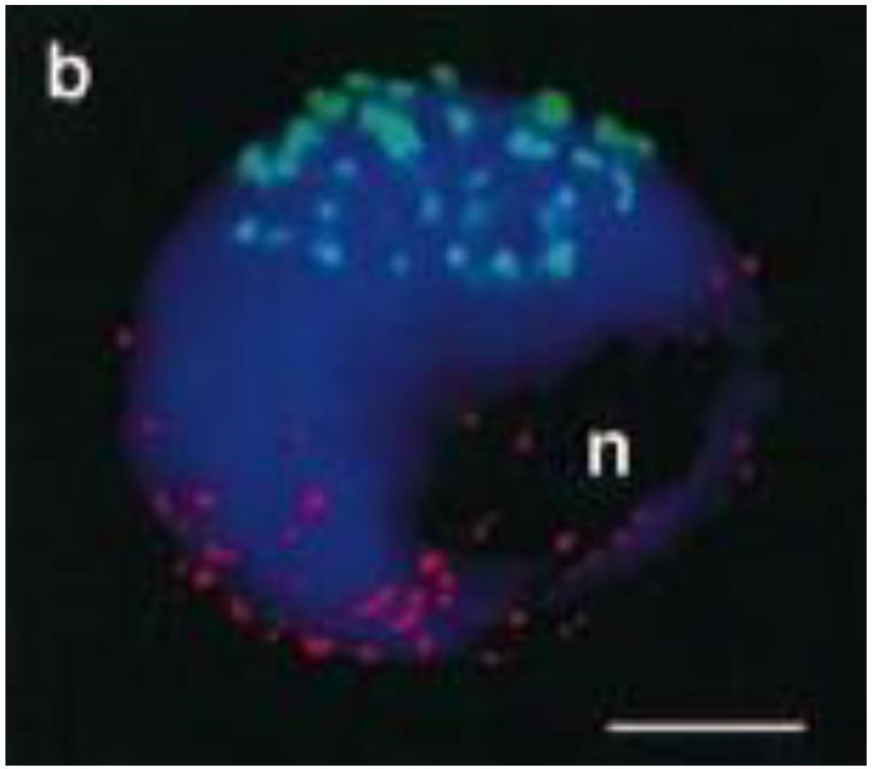




É possível inferir sobre a conformação pelo mapa de contatos

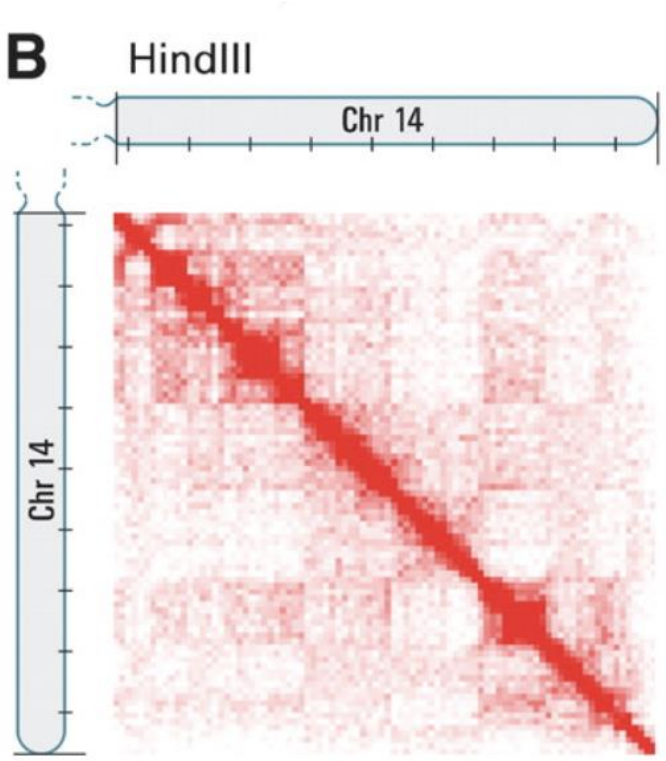




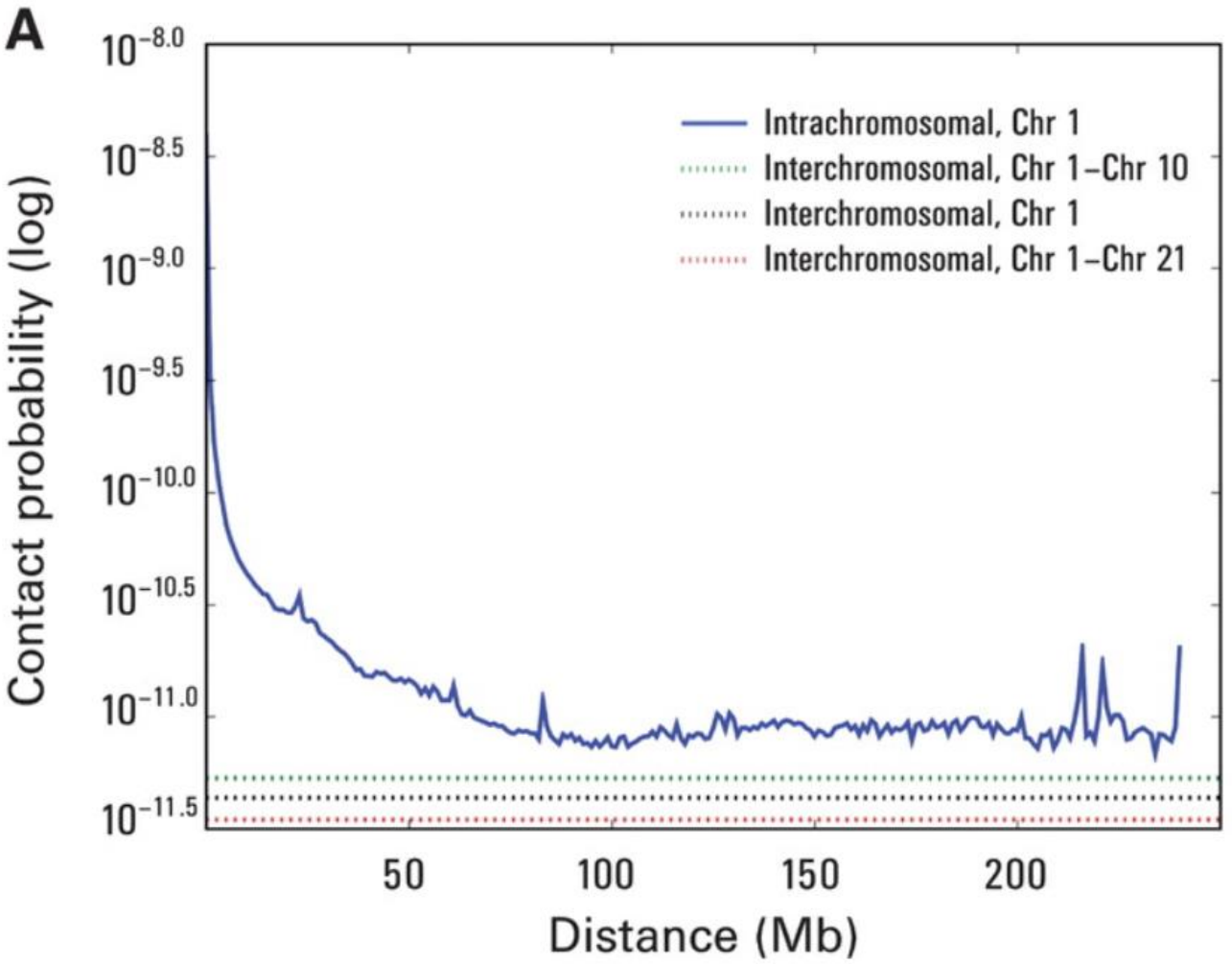


Dong & Jian, 1998.

Hi-C: Frequência ou probabilidade de contatos

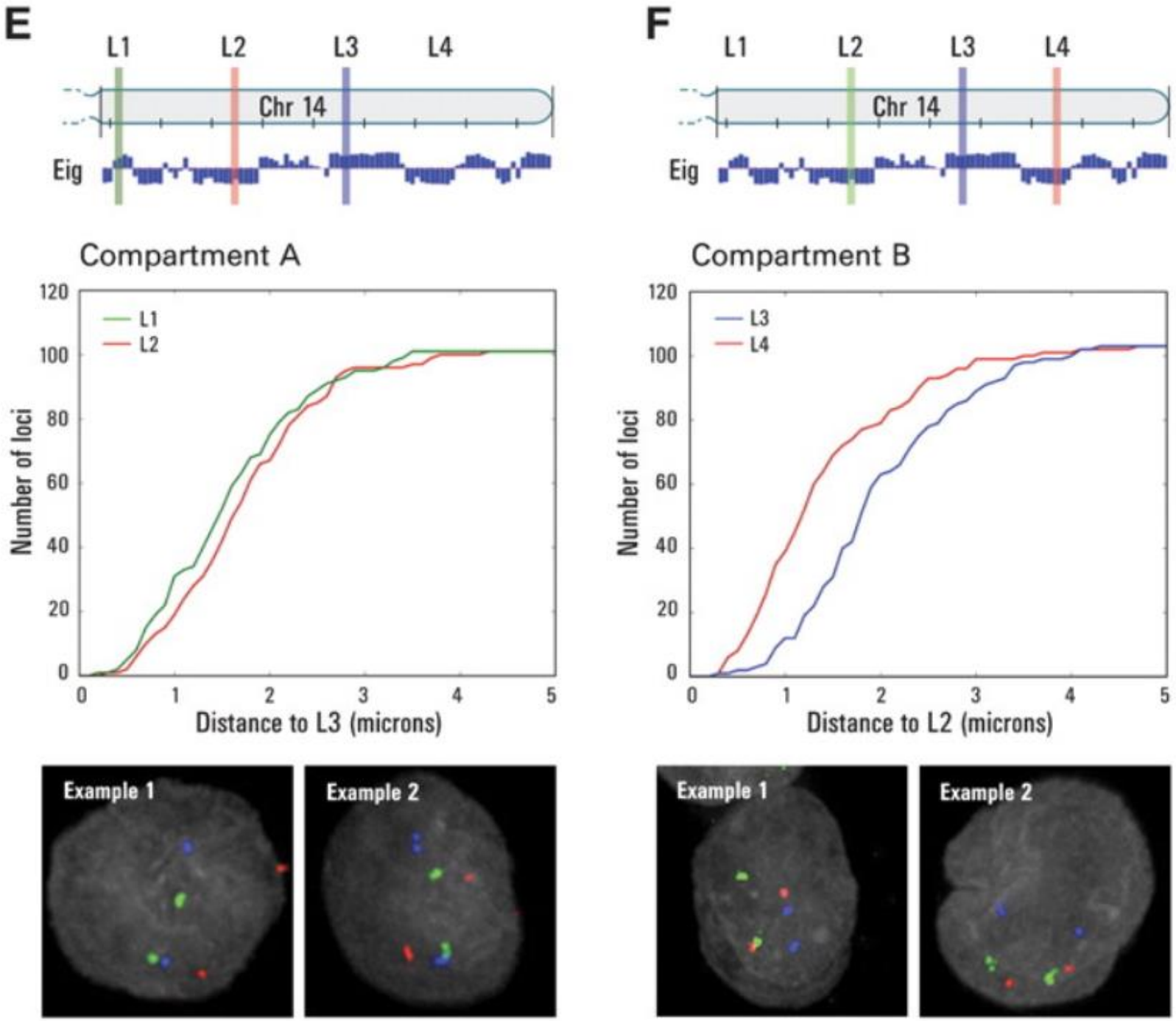


Média de contatos por distância entre sítios ligados



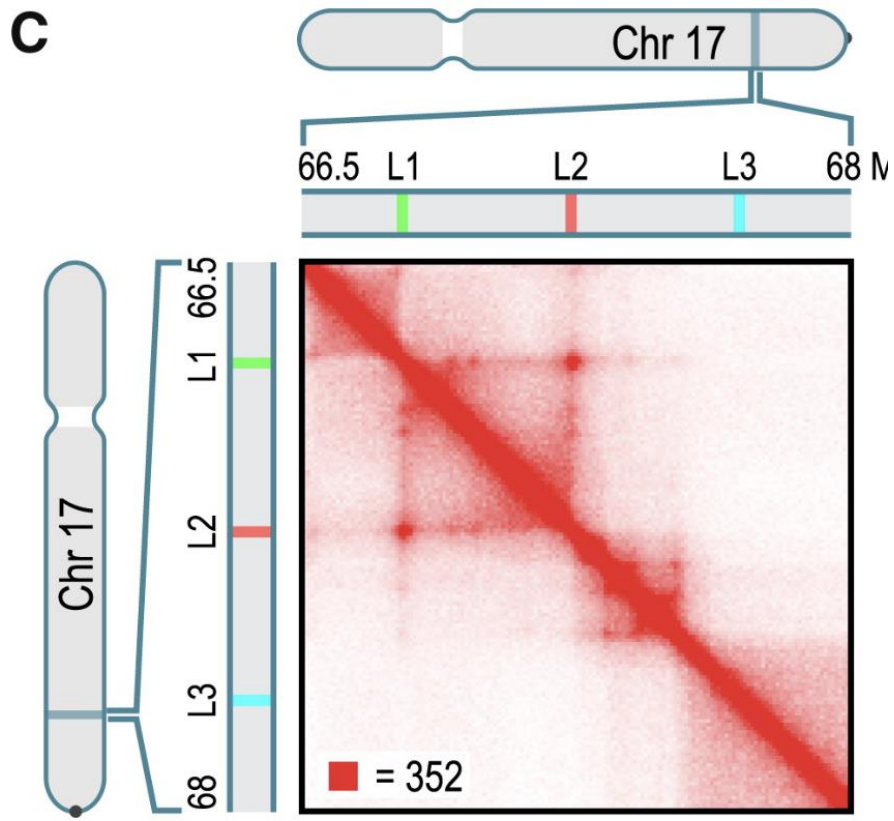
Padrão global, geral de contatos

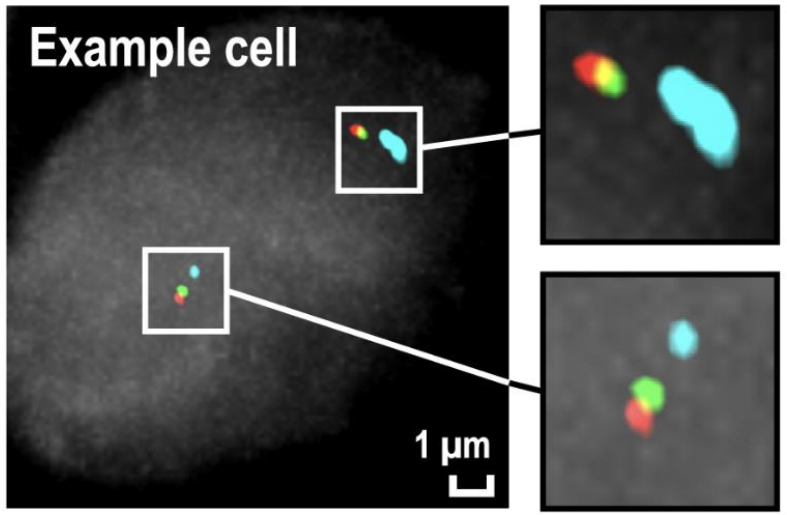
FISH: Distribuição de distâncias entre sítios específicos



Frequência de contatos ~ proximidade entre dois sítios de DNA

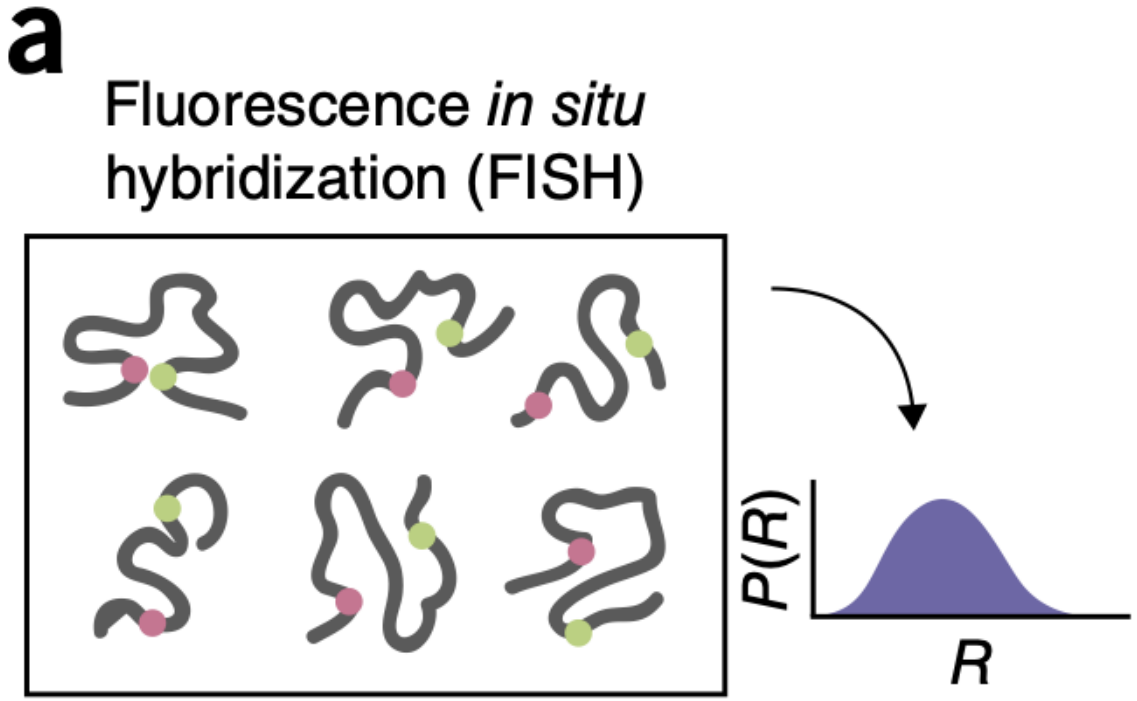
Hi-C & FISH





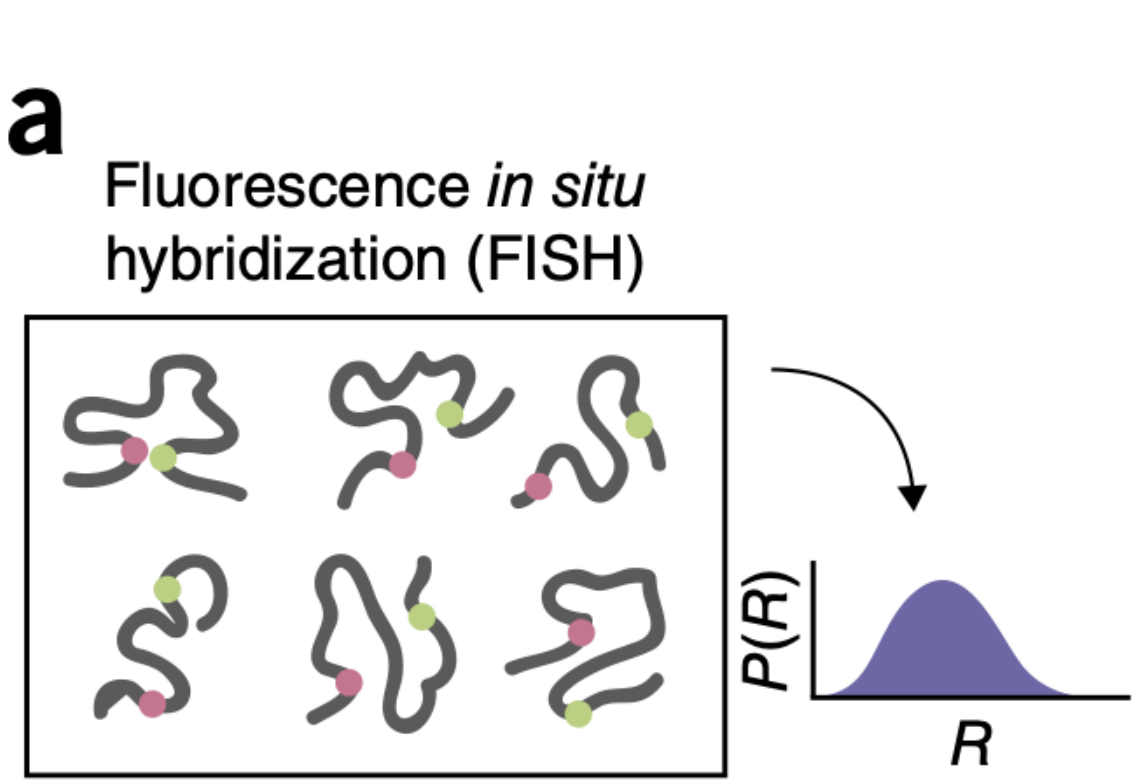
Rao et al, 2014

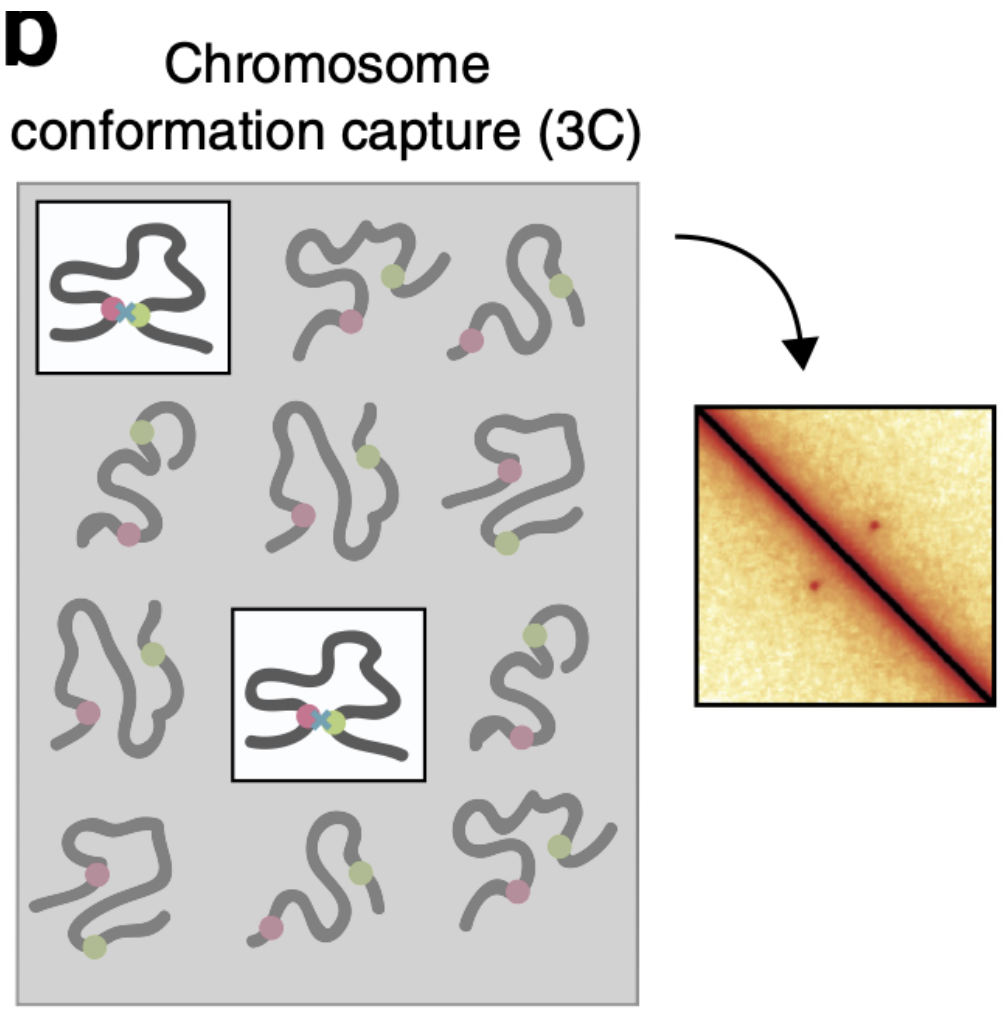
Ensemble de conformações



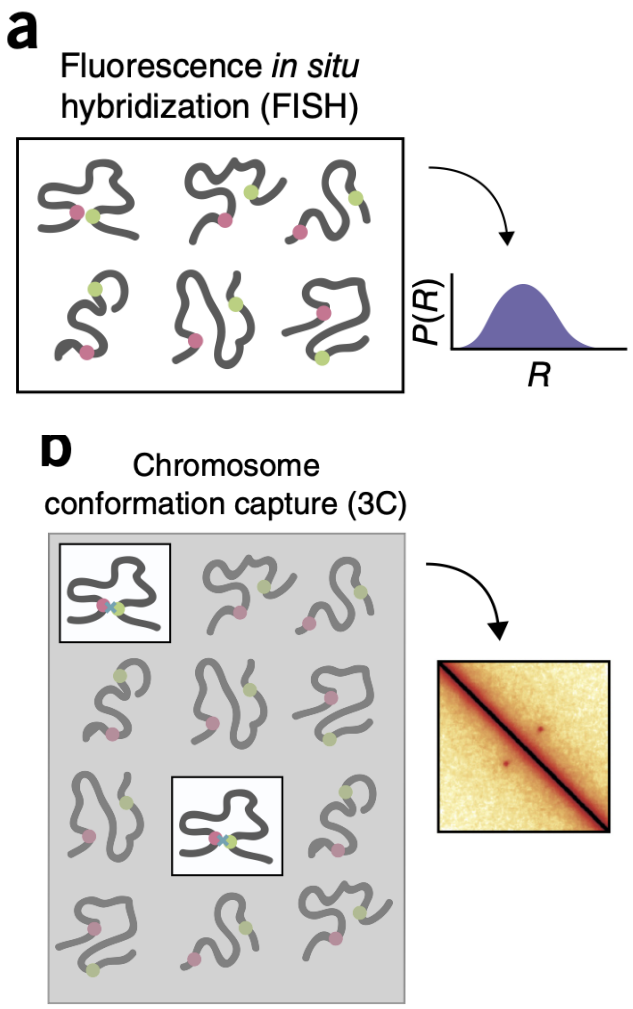
Fudenberg and Imakaev, 2017

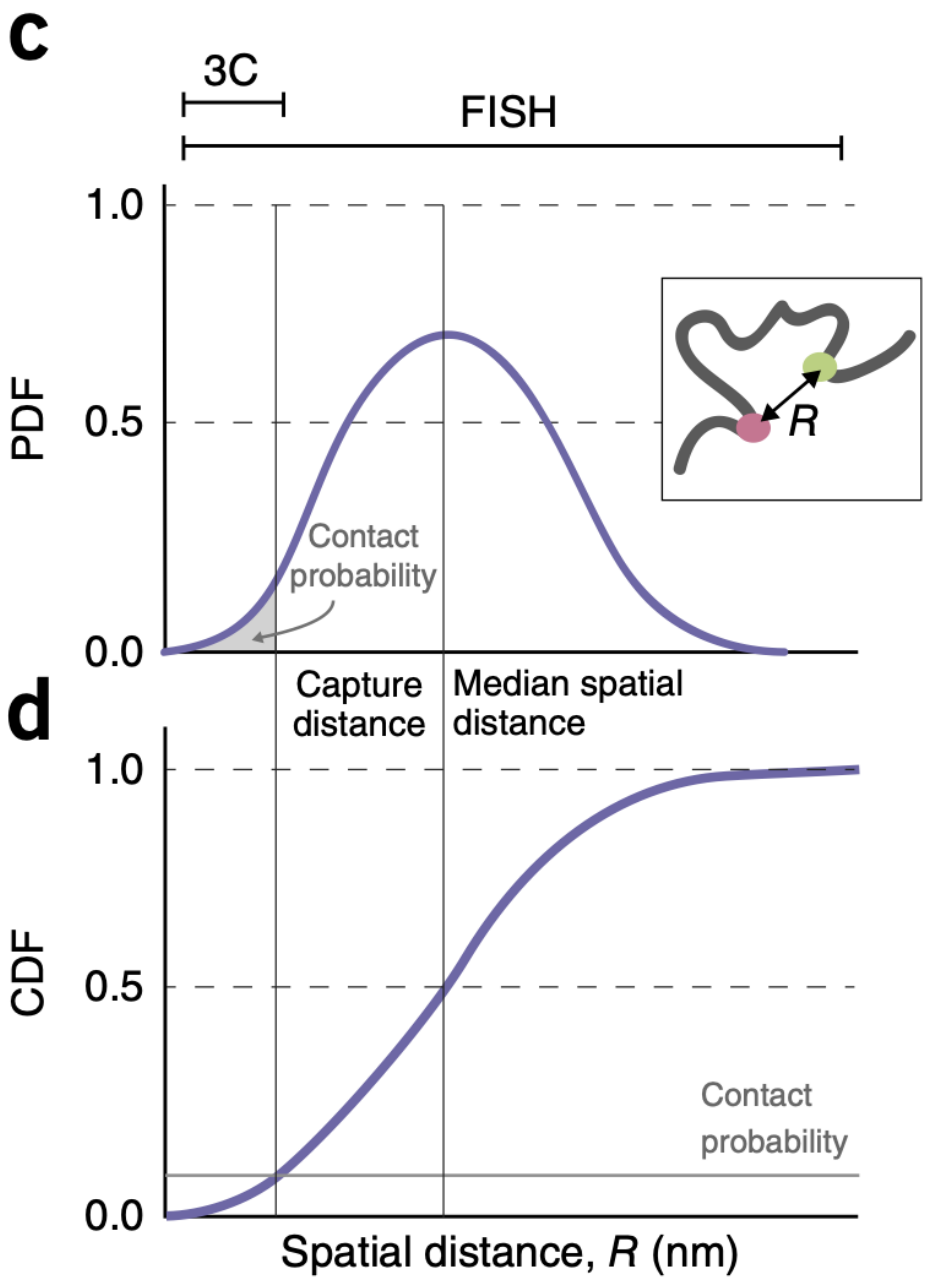
Ensemble de conformações





Ensemble de conformações

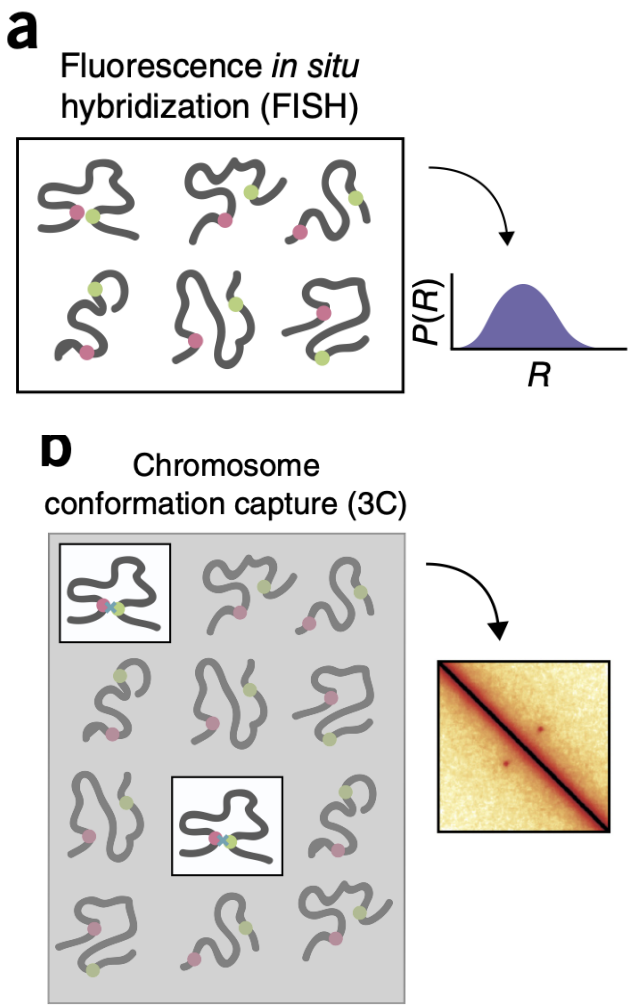


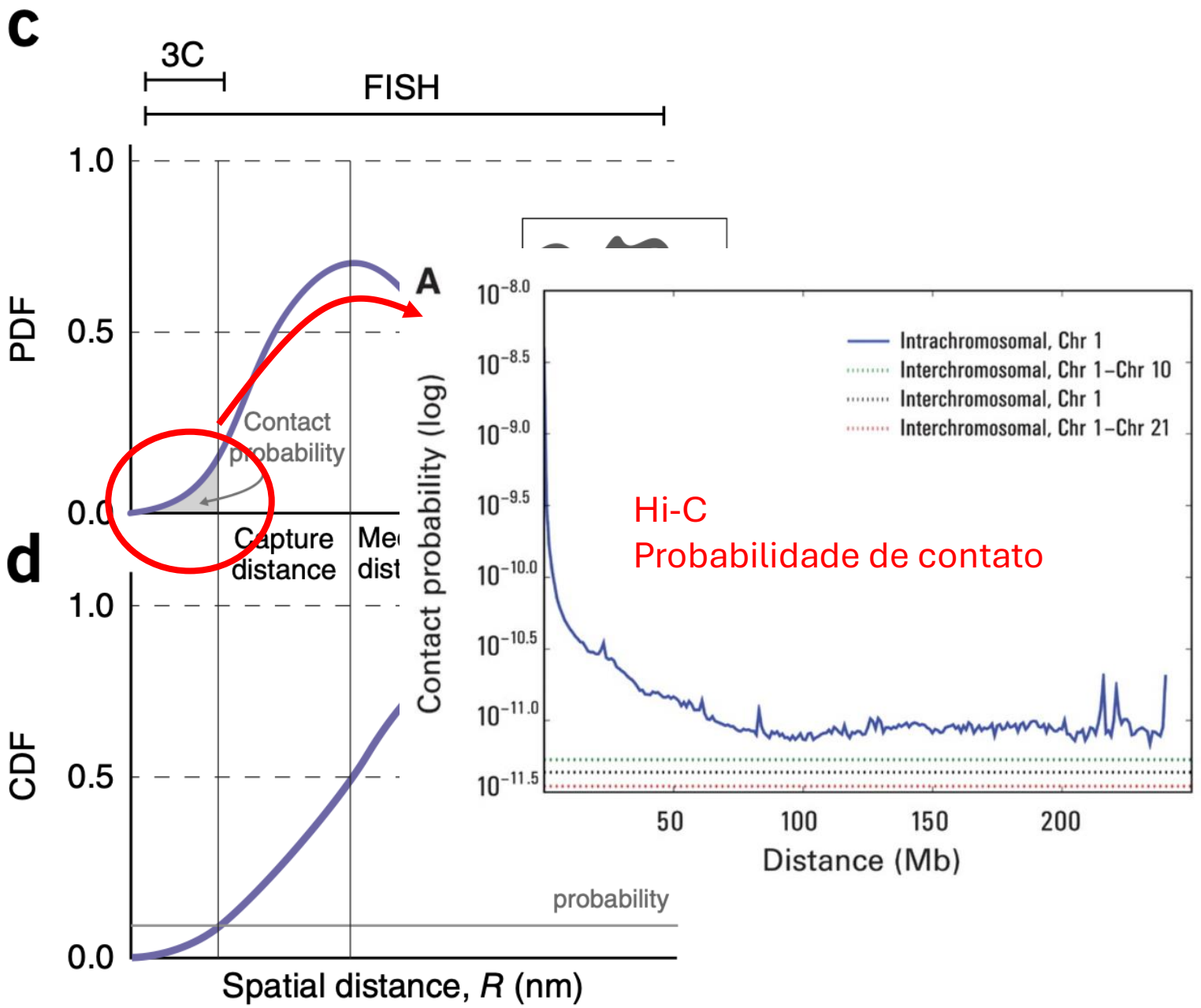


probability density function (PDF)

cumulative distribution function (CDF)

Ensemble de conformações





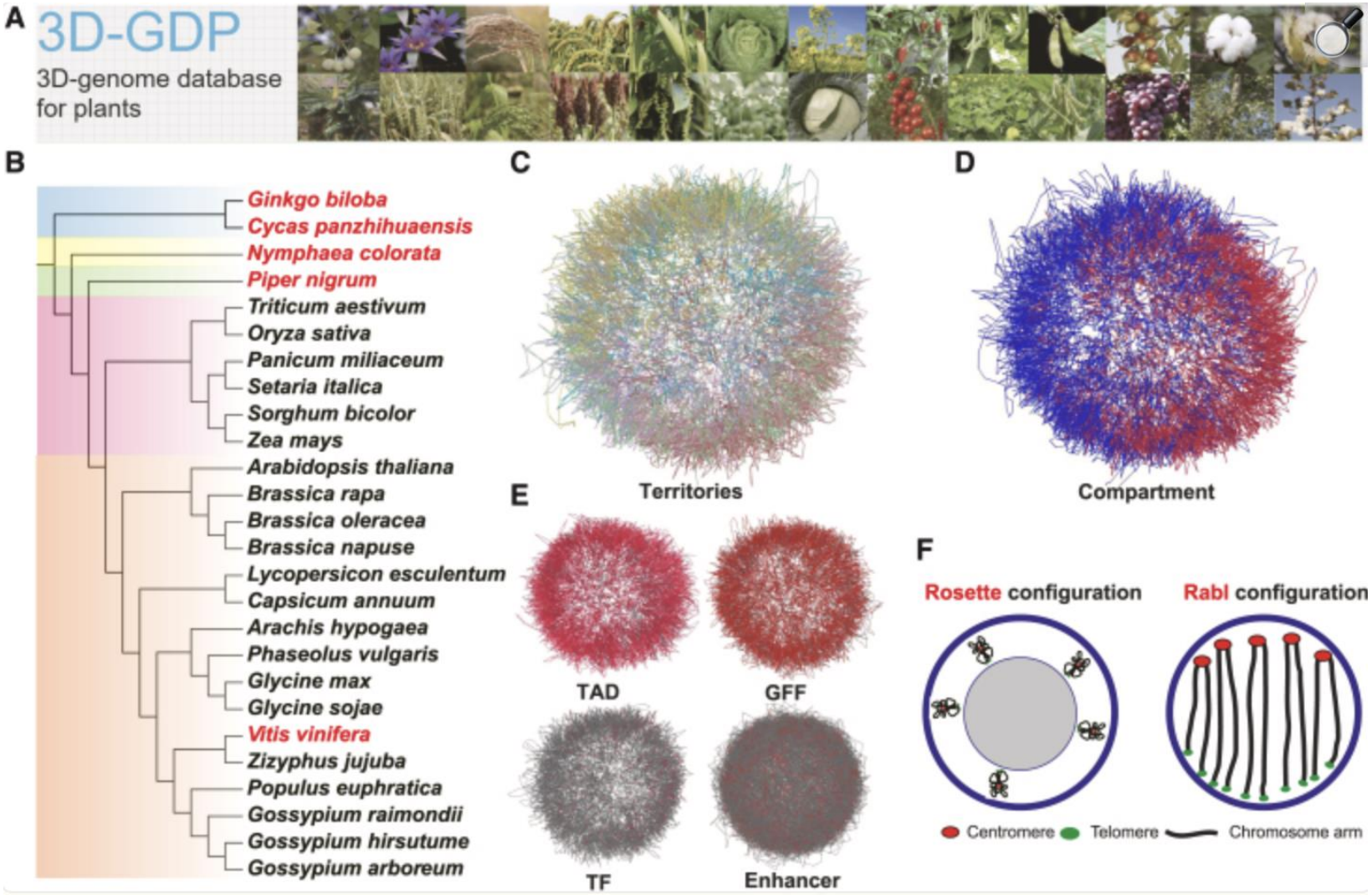
Considerar dois pares de sítios em contextos genômicos diferentes...

É importante considerar a distribuição de distâncias, a dinâmica do DNA.

Modelos computacionais

- Orientado por hipóteses (data driven)
- Orientado por dados (hypothesis driven)

Modelos computacionais por dados

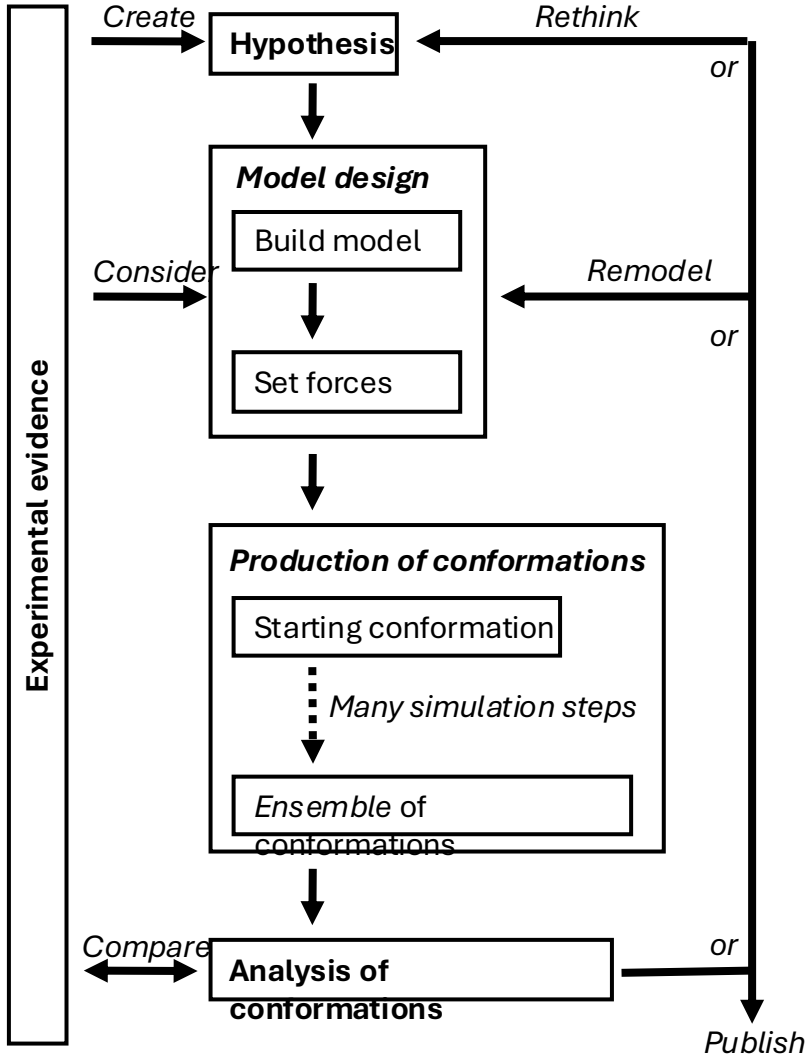


Assumem que
d ~ 1/f

Yang et al, 2025

http://www.3d-gdp.com/

Modelos computacionais por hipótese



Considerations

Build model

- Chromosome region, single chromosome or multiple chromosomes
- Homopolymer or copolymer
- Presence of external particles

Set forces

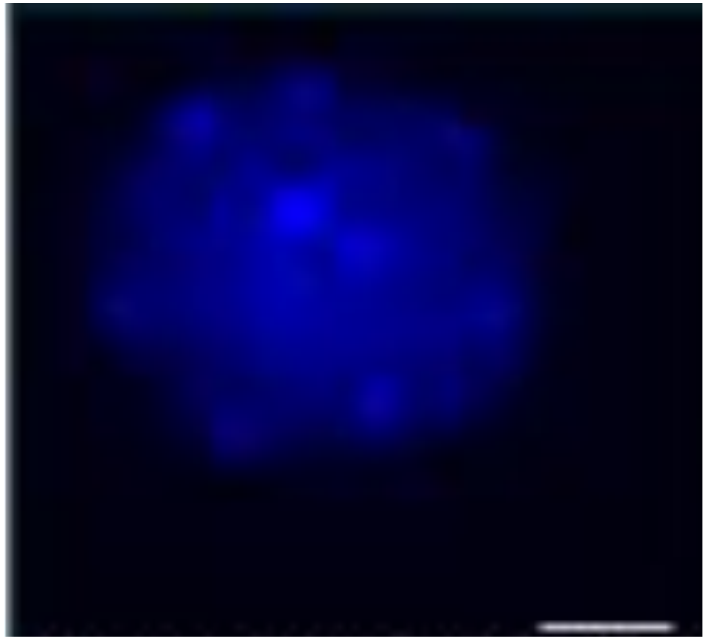
- Intrisic forces
  - Elastic force is mandatory
  - Bending force only if monomer size is smaller than twice the persistence length
  - Non-bonded interaction force may vary with monomer type in a copolymer
- External forces
  - Boundary conditions
  - Attraction to other particles
  - Subjected to function of proteins

Production of conformations

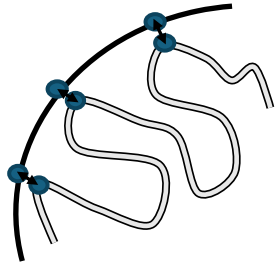
- Starting conformation may be random or predefined
- Different programs are available to perform simulation steps
- Simulation may be of type Monte Carlo or Molecular Dynamics
- If Molecular Dynamics, an equilibration period may be required

Camara and Mascher et al, 2024

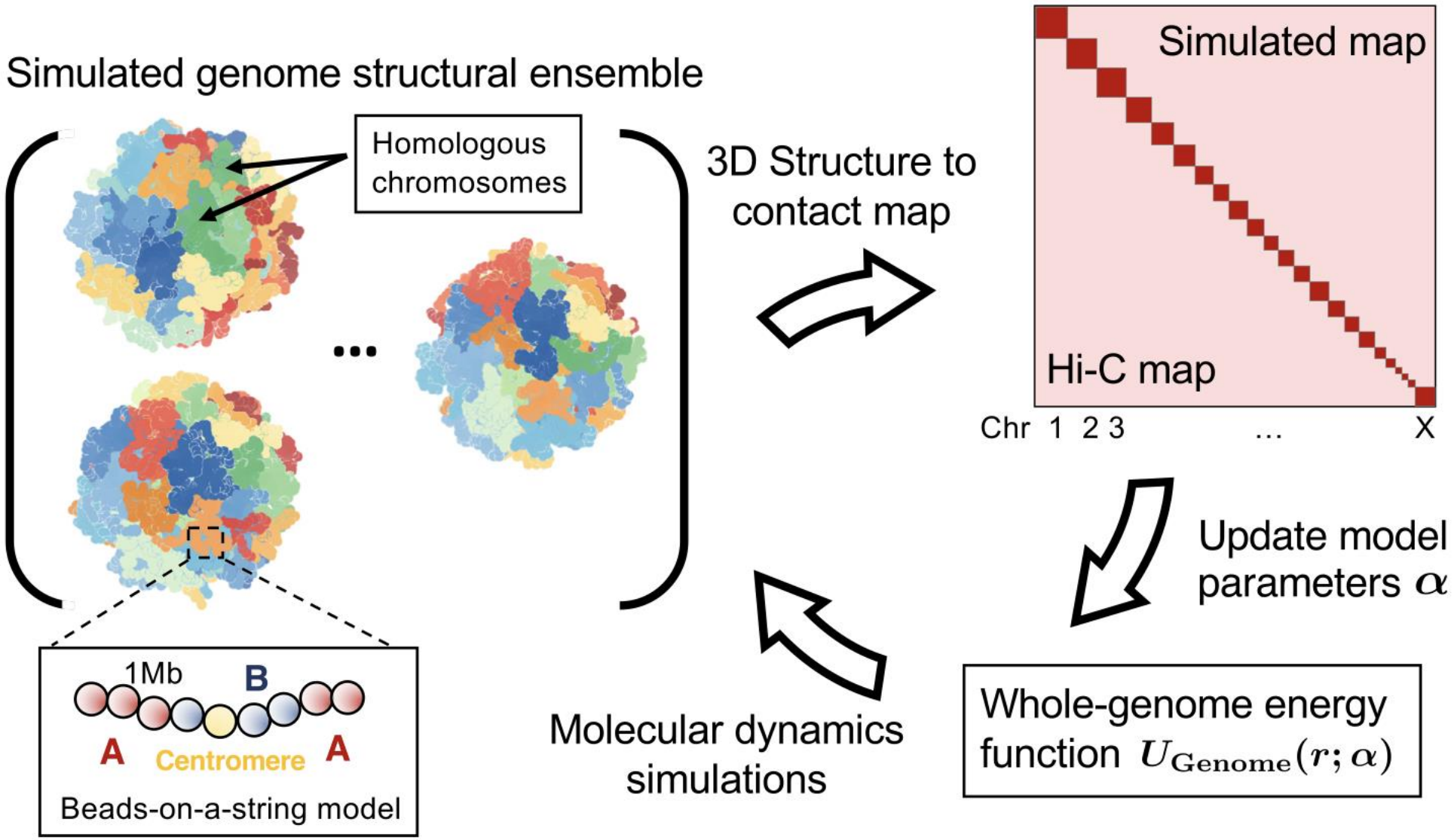
Modelos computacionais por hipótese



Por exemplo:
Acho que histonas de heterocromatina tem afinidade por proteínas de lamina. Por isso que a heterocromatina normalmente fica mais perto da parede do núcleo.
Simular força de atração
Observar consequências: a heterocromatina se distribuiu de forma semelhante ao observado nas imagens de microscopia?



Modelos intermediários (ainda assim, data driven)



Suposições diretas X Ajuste de modelo

Hypothesis-driven vs. data-driven models

Dimension	Hypothesis-driven	Data-driven
Starting point	Theory or research question	Observed data
Main objective	Test an assumption or explanation	Find patterns or make predictions
Strength	Interpretability and focused testing	Flexibility and pattern discovery
Data requirements	Data relevant to a specific hypothesis	Often benefits from large, representative datasets
Causality	Can test causal claims with appropriate design	Patterns alone do not establish causality
Main risk	Bias from incorrect assumptions	Overfitting, biased data, or spurious correlations
Typical output	Evidence supporting or contradicting a hypothesis	Predictions, patterns, classifications, or estimates

*Feito com IA